PROJECT DESIGN PHASE

Solution Architecture

Date	11 October 2026
Team ID	SWTID-2026-7837
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. Architecture Overview

The solution architecture uses native ServiceNow components to automate the Standard Laptop procurement workflow. The architecture connects the Service Catalog and Requested Item with the approval process, Flow Designer, and Catalog Task. After approval, Flow Designer creates the required Catalog Task and assigns it to the Hardware team for configuration.

2. High-Level Solution Architecture

REQUEST LAYER	APPROVAL LAYER	AUTOMATION LAYER	TASK LAYER
Service Catalog Standard Laptop Request ↓ Requested Item	Approval Process Approval Status ↓ Approved Request	Flow Designer Evaluate Approved Request ↓ Create Catalog Task ↓ Assign Hardware Team	Catalog Task Hardware Team ↓ Laptop Configuration

Flow direction: Standard Laptop Request → Requested Item → Approval → Flow Designer Automation → Catalog Task → Hardware Team Configuration.

3. Architecture Layers

Layer	ServiceNow Component	Responsibility
1. Request Layer	Service Catalog	Provides the interface for submitting the Standard Laptop request.
2. Record Layer	Requested Item	Represents the requested Standard Laptop item and carries the procurement context.
3. Approval Layer	Approval Process	Determines whether the request is approved and eligible for the next workflow stage.
4. Automation Layer	Flow Designer	Controls the automated transition from approved request to Catalog Task creation and assignment.
5. Task Layer	Catalog Task	Stores the actionable work item for the Hardware team.
6. Execution Layer	Hardware Team	Receives the assigned Catalog Task and performs laptop configuration.

4. Component Interaction

Component	Receives / Uses	Produces / Performs
Service Catalog	User's Standard Laptop request	Creates the procurement request / Requested Item.
Requested Item	Standard Laptop request details	Provides request context to the approval and automation workflow.
Approval Process	Requested Item / procurement request	Provides approval outcome.
Flow Designer	Approved Standard Laptop request	Creates Catalog Task and assigns it to the Hardware team.
Catalog Task	Approved request context	Provides actionable configuration work for the Hardware team.
Hardware Team	Assigned Catalog Task	Begins Standard Laptop configuration.

5. End-to-End Processing Flow

Step	Architectural Operation	Outcome
1	User submits a Standard Laptop request through the Service Catalog.	Requested Item is created.
2	The request enters the approval process.	Approval status is determined.
3	An approved Standard Laptop request reaches Flow Designer.	Automation begins.
4	Flow Designer creates a Catalog Task.	Hardware configuration task exists.
5	Flow Designer assigns the Catalog Task to the Hardware team.	Task is routed to the responsible team.
6	Hardware team receives the task and performs configuration.	Laptop configuration can begin promptly.

6. Architecture Principles

• Automation: the approval-to-task transition is handled by Flow Designer rather than manual task creation.

• Consistency: approved Standard Laptop requests follow the defined automated workflow.

• Visibility: the Catalog Task provides a clear actionable work item for the Hardware team.

• Reduced manual effort: task creation and assignment are automated.

• Focused scope: the architecture contains only the ServiceNow components required by the stated project functionality.

7. Architecture Summary

The proposed architecture provides a simple ServiceNow-based flow from procurement request to hardware configuration. The Service Catalog captures the Standard Laptop request, the Requested Item and approval process provide the required workflow context, Flow Designer automates task creation and assignment, and the Catalog Task gives the Hardware team a clear work item. This architecture directly supports the project's goals of reducing delays, minimizing manual errors, improving task visibility, and processing approved laptop requests consistently.